

uncentred GRM fit: low rank, diagonal free

genotypes G

GRM
(scaled, uncentred)

L : diag. imputed by
top 3 PCs (rank 4)

S : the diagonal

standardise,
not centred

fit
 $A = L + S$

eigen-
vectors

PCs: PC2+1 vs PC3+1
MXL axis $R^2 = 0.99$

PC3+1

PC2+1

● CEU ● CHB ● MXL ● YRI

